

Finite Open Toda Flow: Lax Isospectrality, Sorting Scattering, and Norming Coordinates

HCS Research Program

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Abstract

We close a convention-complete theorem for the finite open Toda Hamiltonian $H = \frac{1}{2} \sum p_j^2 + \sum_{j < N} e^{q_j - q_{j+1}}$. The Flaschka variables turn its equations into a Lax flow of a positive Jacobi matrix. Conserved traces give global existence and prevent any finite-time edge collision. For simple spectrum, the eigenvalues are fixed and the Moser sorting theorem identifies the two asymptotic velocity orders; the first-component norming weights obey an exact normalized exponential law. For $N = 2$ the full trajectory is a sech/tanh formula. We keep separate the noncompact positive scattering chamber, the phase-compactified isospectral torus, and the repeated-root block boundary. A deterministic ledger audits six rational source rows. This is a source-local integrability result: there is no arithmetic owner, target determinant, or Hilbert–Polya bridge.

1 Model and Lax closure

For $N \geq 2$, use

$$H(q, p) = \frac{1}{2} \sum_{j=1}^N p_j^2 + \sum_{j=1}^{N-1} e^{q_j - q_{j+1}}, \quad \dot{q}_j = p_j, \quad \dot{p}_j = e^{q_{j-1} - q_j} - e^{q_j - q_{j+1}}, \quad (1)$$

where missing endpoint exponentials are zero. Set $a_j = \frac{1}{2} e^{(q_j - q_{j+1})/2} > 0$, $b_j = -p_j/2$, and let L have diagonal b_j and off-diagonal a_j . Let $B_{j,j+1} = a_j = -B_{j+1,j}$. Direct multiplication gives

$$\dot{L} = [B, L], \quad \dot{a}_j = a_j(b_{j+1} - b_j), \quad \dot{b}_j = 2(a_j^2 - a_{j-1}^2), \quad a_0 = a_N = 0. \quad (2)$$

Theorem 1 (Global Lax and Hamiltonian invariants). *Every finite initial state has a solution for all $t \in \mathbb{R}$; all $a_j(t)$ remain positive. The eigenvalues of L and $I_k = \text{tr}(L^k)/k$ are constant, with $H = 4I_2$ and $\sum_j p_j = -2 \text{tr } L$.*

Proof. Since $B^T = -B$, cyclicity gives $\frac{d}{dt} \text{tr } L^k = k \text{tr}(L^{k-1} [B, L]) = 0$. In particular $\text{tr } L^2$ bounds every matrix entry, so the smooth finite-dimensional vector field is complete. Also $a_j(t) = a_j(0) \exp(\int_0^t (b_{j+1} - b_j) ds) > 0$; hence a zero edge is only a boundary limit. Reconstructing $q_j - q_{j+1} = 2 \log(2a_j)$ and $p_j = -2b_j$ proves the Hamiltonian statement. $\square$

2 Simple spectrum and sorting

An irreducible real symmetric Jacobi matrix has simple real eigenvalues. Order them $\lambda_1 > \dots > \lambda_N$. The finite open Toda scattering theorem states

$$b_j(t) \rightarrow \lambda_j \quad (t \rightarrow +\infty), \quad b_j(t) \rightarrow \lambda_{N+1-j} \quad (t \rightarrow -\infty), \quad (3)$$

and $q_j(t) = -2\lambda_j t + c_j^+ + o(1)$ at $+\infty$, with reversed order at the other end. The recurrence proof of simplicity is short: an eigenvector with first component zero has every component zero.

For a normalized eigenvector $v(\lambda_k)$, define $\rho_k = |v_1(\lambda_k)|^2$. The Weyl residues are positive and sum to one, and

$$\rho_k(t) = \frac{\rho_k(0) e^{2\lambda_k t}}{\sum_\ell \rho_\ell(0) e^{2\lambda_\ell t}}, \quad b_1(t) = \sum_k \lambda_k \rho_k(t). \quad (4)$$

Indeed $v_t = Bv$ gives $\dot{\rho}_k = 2(\lambda_k - b_1)\rho_k$, which integrates to (4). The dominant exponent explains the two sorting ends; continued-fraction inversion of the Weyl function supplies the remaining diagonal limits.

3 Closed two-body result and action–angle boundary

For $N = 2$, writing

$$d = \sqrt{(b_1 - b_2)^2 + 4a_1^2}, \quad \alpha = \operatorname{artanh} \frac{b_1 - b_2}{d},$$

gives the exact solution

$$a_1(t) = \frac{d}{2} \operatorname{sech}(dt + \alpha), \quad b_{1,2}(t) = \frac{b_1 + b_2 \pm d \tanh(dt + \alpha)}{2}. \quad (5)$$

Thus the edge is positive at finite time and decays only at the two scattering ends; the eigenvalues are $(b_1 + b_2 \pm d)/2$.

On the simple-spectrum regular set, ordered eigenvalues plus the positive norming simplex $\rho_k > 0, \sum \rho_k = 1$ are inverse-scattering coordinates. After removing the center-of-mass action, logarithmic norming variables furnish local canonical action–angle coordinates. The physical positive isospectral leaf is an open, noncompact scattering chamber, not a compact real Liouville torus. Adjoining complex norming phases compactifies the angles to an $(N - 1)$ -torus; that extension is a geometric boundary description, not a target operator or a periodic Toda claim. More explicitly, in the complexified inverse-spectral chart the nonzero residues have a common $\mathbb{C}^\times$ gauge; fixing their moduli leaves the phase fiber $(S^1)^N/S^1 \simeq \mathbb{T}^{N-1}$ over each simple spectrum. For a direct all- N check, define $r_k(t) = \rho_k(0)e^{2\lambda_k t}$ and the Hankel minors

$$\tau_j = \sum_{|S|=j} \left(\prod_{k \in S} r_k \right) \Delta(\lambda_S)^2, \quad a_j = \frac{\sqrt{\tau_{j-1}\tau_{j+1}}}{\tau_j}, \quad b_j = \frac{1}{2} \partial_t \log \frac{\tau_j}{\tau_{j-1}}.$$

Dominant top/bottom subsets prove the two sorting ends exactly; the ledger uses the equivalent normalized-residue form.

4 Reproducible ledger

The receipt uses six rational a, b rows (three each for $N = 2, 3$), 256 RK4 steps per unit time, and 90-decimal arithmetic. It stores all 30 states at $t = -2, -1, 0, 1, 2$, every trace invariant, sorted eigenvalues, positivity and drift. Fifteen $N = 2$ rows compare (5) directly; nine $N = 3$ rows compare (4) at $t = -1, 0, 1$. Six $T = 8$ endpoint rows are finite diagnostics of (3), not exact limits. Representative initial spectra are

case	N	(a_j)	(b_j)
N2 reference	2	(1)	(1/2, -1/2)
N2 asymmetric	2	(3/4)	(1/3, -1/6)
N3 symmetric	3	(1, 1)	(1, 0, -1)
N3 generic	3	(1/2, 3/2)	(1/2, -1/4, 1/3)

The independent checker reconstructs the ODE and matrix spectrum without importing the producer. SymPy checks 25 identities; canonical replay is byte-identical and the hostile suite rejects 22 of 22 repaired mutations.

5 Degenerate face and strict Route-A boundary

At $a_j = 0$, L splits into blocks and regular simplicity/action–angle coordinates stop applying. The explicit $N = 3$ boundary $L = \operatorname{diag}(0, 0, 1)$ has characteristic polynomial $x^2(x - 1)$: the repeated root is a block degeneracy, not a positive Jacobi spectrum. The finite characteristic polynomial $\det(\lambda I - L)$ is source-local and is not an orbit zeta or Fredholm determinant.

The strict evaluator tuple is

$$(\text{AO_FAIL}, \text{A1_FAIL}, \text{A2_FAIL}, \text{A3_FAIL}, \text{A4_FORMAL_HINT}),$$

with `ROUTE_A_REJECTED` and `route_b_invocation_allowed=false`. The open chain has no rational-prime owner, primitive periodic repetition law, target divisor, or natural Hilbert–Polya lift. Its Hamiltonian scattering theorem remains a substantive positive result.

References

- [1] H. Flaschka, The Toda lattice. II. Existence of integrals, *Physical Review B* 9 (1974), 1924–1925. DOI: 10.1103/PhysRevB.9.1924.

- [2] J. Moser, Finitely many mass points on the line under the influence of an exponential potential—an integrable system, in *Dynamical Systems, Theory and Applications*, Lecture Notes in Physics 38, Springer (1975), 467–497. DOI: 10.1007/3-540-07171-7_12.
- [3] C. Tomei, The topology of isospectral manifolds of tridiagonal matrices, *Duke Mathematical Journal* 51 (1984), 981–996. DOI: 10.1215/S0012-7094-84-05144-5.

Proof and audit supplement

The first-component identity behind (4) is exact. If $Lv_k = \lambda_k v_k$ and $\dot{v}_k = Bv_k$, then $(\dot{v}_k)_1 = a_1(v_k)_2 = (\lambda_k - b_1)(v_k)_1$; squaring and normalizing gives $\dot{\rho}_k = 2(\lambda_k - b_1)\rho_k$. The positive residues and ordered eigenvalues determine the Weyl rational function $m(z) = \sum_k \rho_k / (z - \lambda_k)$. Its Stieltjes continued fraction uniquely recovers every positive Jacobi coefficient, so concentration of the residues at the two time ends yields the complete sorting statement rather than only the b_1 limit.

The finite receipt is a control on the implementation, not the proof of the limits. Across all stored times, the largest trace-invariant drift is 2.79×10^{-10} , the largest spectral drift is 1.17×10^{-10} , the largest two-body formula error is 4.55×10^{-11} , and the largest norming-law discrepancy is 1.71×10^{-10} . Weaker links converge more slowly at $T = 8$, so their nonzero endpoint errors are retained. The producer-independent checker makes 596 assertions; all finite rows remain strictly inside the positive chamber.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic, target-zero, determinant matching, or Hilbert–Polya claim. **Data and code.** All rows and audits accompany HCS-C230. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checks displayed formulas and metadata. This is not external peer review.